



MAO SEDTHA

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ABOUT ME

I am a driven senior data science student's at the Institute of Technology of Cambodia with a strong passion for Artificial Intelligence, Robotics, and real-world problem solving. Strive toward creating machine learning models computer vision systems and object detection project. Passionate about developing data driven solutions to tackle complex problems and improve the human condition. I thrive in research environments, value teamwork, and am eager to contribute to innovative projects that advance technology for social impact.

WORK EXPERIENCE

Assistant Data Analyst Internship / The SEARLE Company Ltd **September - October 2024**

- Analyzed data on medicine imports from various countries, providing actionable insights to support decision-making.
- Designed and developed dynamic dashboards and reports in Power BI to visualize import trends, country-wise contributions, and key performance indicators.
- Collaborated with team members to interpret findings and align visualizations with organizational objectives.

VOLUNTTERING

IDP Education **15 March 2024**

- Volunteered as an interpreter to support international students during university sessions and enrollment
- Helped bridge communication with staff, ensured accurate translation, and contributed to a welcoming, inclusive environment

ACADEMIC PROJECTS

AI-Powered Admission Intelligence Platform **January 2025**

- Designed and built an end-to-end AI system to support student admission decision-making
- Developed a RAG-based chatbot using LangChain + vector database to answer program-specific queries from structured and unstructured data
- Implemented ML models to predict student success/admission likelihood, improving decision support
- Built scalable backend APIs using FastAPI for model serving and chatbot interaction
- Designed system architecture integrating LLM + ML + database + API layer

AI Chatbot with Structured Knowledge (RAG System)

January 2025

- Built a Retrieval-Augmented Generation (RAG) chatbot using structured datasets
- Designed pipeline for data ingestion → embedding → retrieval → LLM response
- Improved answer relevance using prompt engineering and retrieval tuning

Autonomous Driving Agent (Reinforcement Learning)

January 2025

- Developed a self-driving agent in CARLA using reinforcement learning
- Implemented Proximal Policy Optimization (PPO) with Stable-Baselines3
- Designed custom environment, reward functions, and training pipeline
- Optimized training performance (timesteps, FPS, environment stability)
- Integrated sensor inputs (camera / state) for decision-making

Face Classification and Detection Project

January 2025

- Developed a deep learning-based system using PyTorch for real-time face detection and recognition.
- Utilized pre-trained models (e.g., Faster R-CNN, FaceNet) and custom datasets with annotated images.
- Implemented image preprocessing, face embedding, and classification to accurately identify individuals across multiple classes.

EDUCATION

Institute of technology of Cambodia

- Major : Data Science

Anlong veng high school

2021 - Present

ADDITIONAL INFORMATION

2015 - 2021

Technical Skills:

- Programming Languages: Python, C++, C, Java, SQL
- Frameworks & Libraries: PyTorch, TensorFlow, OpenCV, Seaborn, Matplotlib
- Machine Learning Skills: Supervised & Unsupervised Learning, Classification, Object Detection

Languages:

- Khmer (Mother Tongue)
- English (Intermediate)
- French (Beginner)

Certifications:

- Data Science Camp (UNESCO UNITWIN in Cambodia March-2023)
 - SAP Analytics Cloud and Build Apps training session (Asian data science explorers April-2024)
 - ACTSmart Hackathon 2025 (Held on July 12 - 13, 2025)
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